Beyond Aggregate Accuracy: A Leakage-Aware and Threshold-Sensitive Evaluation of RGB Wildfire Detection on D-Fire

Abstract—Computer-vision methods for wildfire monitoring are often summarized by average precision or image-level accuracy. Neither measure, by itself, indicates how frequently a detector misses localized fire instances or how the alert burden changes when its confidence threshold is adjusted. This paper evaluates a single-stream RGB YOLOv8n detector on the D-Fire dataset and reports both object-level and image-level outcomes. The held-out test partition contains 3,264 images and 3,990 annotated object instances. Frames were partitioned using 32-frame numeric-stem groups; this procedure prevents the same constructed group from crossing splits, but it is only an approximation to source-video separation because video identifiers were unavailable. The detector obtains box-level mAP@0.5 of 0.719 and mAP@0.5:0.95 of 0.385. At the reported operating point, aggregate box-level precision, recall, and F1 are 0.739, 0.664, and 0.700, whereas image-level precision, recall, and F1 are 0.971, 0.945, and 0.958. The image-level accuracy is 0.955 (provided 95% bootstrap interval: 0.948–0.962). Class-specific fire recall is lower than smoke recall (0.572 versus 0.720). An 18-point confidence sweep further shows that fire recall is 0.528 when 7 of 1,504 negative images (0.47%) contain a fire detection, and 0.698 when 27 negative images (1.8%) do. Mean and 95th-percentile per-image latency are 11.7 and 17.1 ms on an NVIDIA RTX 5070 Laptop GPU at 640-pixel input and FP32. These are benchmark measurements, not field-level alarm rates. The results support a reproducible, threshold-aware evaluation of one RGB detector; they do not establish thermal or IIoT fusion benefits, false alarms per camera-hour, early-warning lead time, or embedded-device performance. The manuscript identifies these distinctions and proposes the additional experiments needed before operational claims can be made.

Index Terms—Wildfire detection, smoke detection, object detection, YOLOv8, D-Fire, confidence threshold, false alarms, dataset leakage, operational evaluation.

1 Introduction

Wildfire detection from images is a difficult recognition task because the visual evidence changes with distance, illumination, weather, camera position, and the stage of combustion. Flames may be small relative to the image or partly obscured by vegetation. Smoke can be diffuse, low contrast, or visually similar to cloud, fog, dust, steam, or industrial emissions. A system that raises alerts early must therefore balance two different errors: failing to identify a real event and repeatedly alerting on non-fire scenes.

The evaluation of such systems requires careful definition of the unit of analysis. At the image level, a frame may be counted as positive if it contains at least one correct detection. This does not establish that every fire or smoke instance was localized. Conversely, object-level measures assess localization and classification for annotated boxes, but do not directly state how many camera images or operational time intervals will produce nuisance alerts. Both views are relevant, but they answer different questions. A high image-level score should not be interpreted as a high probability of correctly localizing every object, and a per-image false-positive proportion should not be interpreted as alarms per hour in a video stream.

A second issue concerns the independence of training and test data. Fire datasets may contain adjacent video frames or multiple images from the same event. If correlated frames are distributed across training and test partitions, the measured score may partly reflect familiarity with a scene rather than generalization to an independent event. This study uses a group-aware partition constructed from consecutive numeric-stem buckets. The resulting groups do not cross the supplied train, validation, and test partitions. However, the dataset package used for this work does not provide source-video identifiers for all frames. The split should therefore be understood as an approximation, not proof of event-level independence.

The experimental archive supplied for this study contains a trained YOLOv8n RGB detector, a D-Fire benchmark result, a confidence-threshold analysis, false-positive examples, training records, and a number of supplementary design artifacts. Some supplementary material describes a larger RGB/long-wave-infrared (LWIR)/industrial Internet-of-Things (IIoT) pipeline. Those materials do not change the experimental scope of this paper. The evaluated checkpoint is single-stream RGB. A grayscale proxy was used in one development pathway for the thermal channel, full dual-stream training was not completed, and the field-simulation tables are explicitly synthetic. They are not treated as evidence for multimodal superiority or field performance.

The study addresses three questions. First, what object-level and image-level performance is measured for the trained RGB detector on the documented D-Fire test partition? Second, how does fire-instance recall change when the confidence threshold is varied against a false-positive budget defined over negative images? Third, what claims about computation and deployment are supported by the measured inference latency, and which remain outside the available evidence? The purpose is not to propose a new detector architecture. Rather, it is to provide a careful account of the evidence in the supplied experiment and to make the limits of that evidence visible alongside the headline metrics.

The contributions are as follows. (1) The paper documents a leakage-aware grouping procedure and clearly distinguishes the resulting numeric-stem groups from verified source-video groups. (2) It reports object-level and image-level outcomes separately, including the class-specific difference between smoke and fire detection. (3) It presents a threshold analysis in which fire recall is paired with the fraction of negative images producing a fire detection. (4) It separates measured RGB benchmark results from synthetic

replay tables and unmeasured components of the proposed multimodal system. These contributions are intended to improve interpretability and reproducibility rather than to imply readiness for deployment.

2 Related Work and Research Context

Research on fire and smoke recognition has developed from hand-designed visual features and video analysis toward convolutional object detectors, aerial platforms, and temporal or transformer-based models. The literature does not form a single directly comparable leaderboard. Studies use different image sources, event distributions, label definitions, sensor modalities, split protocols, confidence thresholds, and reporting conventions. The discussion below therefore summarizes methodological context rather than ranking the cited systems by their published scores. Each study is cited once, in the order in which it is introduced.

2.1 Datasets and temporal evidence

Dewangan et al. introduced FIgLib and SmokeyNet as a dataset and model for real-time wildland-fire smoke detection. The work is relevant because it treats smoke detection as a temporal problem rather than relying solely on a single isolated image. Its sequence-based design motivates the distinction between frame-level object recognition and the later decision to raise an event alert. In the present study, by contrast, the test measurements are image based; the available material does not provide a continuous, timestamped video evaluation from which event-level detection time could be calculated [1].

Ghali et al. investigated deep-learning and transformer approaches to wildfire detection and segmentation in UAV imagery. This line of work addresses aerial scene geometry and the localization of small or visually sparse fire regions. It also illustrates that segmentation and detection are related but not identical tasks. A segmentation score cannot be compared directly with box mAP unless the task definition and evaluation protocol are aligned. The current study consequently reports only the detection and image-level measures available for the RGB model [2].

Zhao et al. combined saliency detection with deep learning for wildfire identification in UAV imagery. Saliency methods seek visually distinctive regions before or alongside classification. Such work is pertinent to the general challenge of identifying a small fire signal against a complex landscape, but its reported result belongs to the data and evaluation protocol used by the authors. It is not treated here as a control model because the supplied experiment does not include a common reimplementation or predictions on the same held-out partition [3].

2.2 Object detection and benchmark comparison

Gonçalves et al. evaluated YOLO-based models for smoke and wildfire detection in ground and aerial images. The combination of viewpoints is important because the same class can appear at substantially different scales and under different backgrounds. Results obtained from a mixture of ground and aerial imagery can inform model design, but cannot establish performance on a different test distribution without matched data and a shared evaluation protocol. This observation supports the decision to describe the current measurements as D-Fire-specific [4].

Catargiu et al. compared YOLO-type detectors on an open fire-and-smoke dataset. A comparative benchmark is useful because it places multiple detector families under a common experimental setting. Even in such a benchmark, the resulting ranking remains conditional on the dataset, annotation procedure, selected model versions, input resolution, thresholding, and split construction. The present study does not claim superiority over those or other published systems: no matched comparison was supplied, and published numbers from different test protocols are not interchangeable [5].

El-Madafri et al. emphasized dataset diversity, representativeness, and a multi-task learning approach for wildfire detection. This work is relevant to a central limitation of image-based fire benchmarks: a large number of images does not necessarily represent the diversity of weather, geography, fire stage, camera optics, season, or negative look-alike scenes that a field system will encounter. The present D-Fire test set is therefore treated as a benchmark sample, not as a representative sample of every deployment environment [6].

2.3 Architectural approaches for aerial smoke detection

Mukhiddinov et al. developed an optimized YOLOv5-based system for detecting wildfire smoke in UAV images. Their study is part of the work that adapts one-stage detection to aerial smoke, where small target scale and background texture can be important. The current model uses YOLOv8n and a different dataset split; the two systems are not compared numerically because their underlying test examples and evaluation details are not shared [7].

Saydirasulov et al. proposed an improved YOLOv8 approach for UAV smoke imagery. The article illustrates that detector modifications, including feature extraction and attention mechanisms, are often evaluated within a specific UAV image setting. Such architectural changes may improve an in-domain score, but this does not by itself establish cross-camera or cross-region robustness. The results in this paper accordingly concern the trained baseline checkpoint and do not attribute performance to any unmeasured modified detector [8].

Bahhar et al. studied a staged YOLO model and an ensemble CNN for wildfire and smoke detection. A staged design can separate classification and localization decisions, while an ensemble may combine complementary outputs. However, added components also create further requirements for latency, calibration, and failure analysis. Since the present archive contains no matched staged-model experiment, this paper limits its claim to the single RGB detector that was actually trained and evaluated [9].

Chen et al. proposed RepVGG-YOLOv7 for fire and smoke detection. This work provides a further example of modifying a one-stage detector for the visual characteristics of fire and smoke. Published accuracy or mAP values can be useful context, but they should not be interpreted as a

controlled comparison when the datasets, train-test separation, class balance, object-size distribution, and confidence settings differ. For this reason, the present Results section does not juxtapose headline scores as if they came from a common experiment [10].

Huang et al. addressed small-target forest-fire smoke detection using an end-to-end deformable transformer. Small targets are particularly important in early observation because distant smoke or flame regions may occupy few pixels. Nevertheless, small-target model performance depends on input resolution, annotation granularity, and the distribution of scales in the test set. The current evaluation does not include a size-stratified analysis, so it cannot determine whether the lower fire recall is concentrated among small objects [11].

2.4 Temporal, classical, and surveillance-based methods

De Venâncio et al. proposed a hybrid fire-detection method based on spatial and temporal patterns. The combination is conceptually relevant to the operational gap between a frame prediction and a stable alarm: temporal evidence can potentially suppress transient detections. The current archive has a contextual model trained on real embeddings, but only its validation window accuracy is supplied; there is no end-to-end temporal alarm-rate evaluation on the held-out video stream. Its score is therefore not used as a deployment result [12].

Gubbi et al. examined video smoke detection using wavelet features and support vector machines. This earlier work reflects a broader research history in which smoke detection relied on hand-designed appearance or frequency features and video-level evidence. Contemporary deep models differ substantially in representation learning, but they face a continuing evaluation challenge: smoke-like backgrounds and the selection of an operating threshold remain important regardless of model family. This paper evaluates a modern detector, but does not conclude that its score supersedes prior methods under a shared test [13].

Yuan developed video-based smoke detection using histogram sequences of local binary pattern features and their variance pyramids. The use of spatial texture together with sequences highlights the role of temporal consistency and appearance changes in video smoke analysis. Its processing assumptions and image resolution differ from the present object-detection setup. The comparison is consequently conceptual: both bodies of work motivate reporting false alarms and processing constraints, while the supplied RGB evaluation provides only per-image latency and negative-image alarm proportions [14].

Gunay et al. investigated video-based wildfire detection at night. Nighttime monitoring introduces conditions that can be unlike daylight smoke imagery, including bright point sources, changing illumination, and low signal-to-noise backgrounds. The source archive identifies night and flare ambiguity as one possible influence on fire detection in D-Fire, but no controlled daylight-versus-night subgroup metrics are supplied. Therefore, the present paper treats nighttime performance as an unresolved domain question rather than inferring it from aggregate test results [15].

Zhou et al. studied wildfire smoke detection using local-extremal-region segmentation and surveillance. Long-distance smoke surveillance emphasizes region proposals, temporal movement, and the possibility of false detections from background structures. The supplied error examples in the current project include clouds and man-made structures, which are qualitatively consistent with the general difficulty of separating smoke or fire appearance from background. These examples are descriptive only; they do not constitute a complete or independently coded error taxonomy [16].

2.5 Synthesis and specific gap

Across the studies summarized above, three methodological issues are especially relevant to the present work. First, benchmark results depend on dataset composition and partitioning. Second, per-frame recognition and event-level alerting are distinct outputs. Third, the operational significance of a detector depends on the selected threshold and on the temporal frequency of errors. The available experiment permits analysis of the first two issues only in a limited form and provides a confidence sweep for the third at the negative-image level. It does not permit a controlled architecture comparison, a verified source-video split, or calculation of alarms per camera-hour. The specific gap addressed is therefore an evidence-bounded evaluation of object and image metrics together with a threshold trade-off, accompanied by a precise statement of the unmeasured system claims.

3 Materials and Methods

3.1 Study design and scope

This work is a retrospective analysis of an existing trained checkpoint and its associated benchmark artifacts. No new participants, field cameras, controlled burns, or sensor deployments were introduced for the present analysis. The principal outcome is detector performance on the held-out D-Fire test partition described in the experiment archive. The analysis was designed to avoid combining measurements from incompatible sources. In particular, real test-set results are reported separately from synthetic replay tables and from training-only contextual model metrics.

The experimental materials include a detector checkpoint, metric files, a test prediction table, confidence-threshold outputs, a dataset audit, training history, and latency traces. The archive documents that the reported detection metrics and image-level metrics are measured on the held-out test images. The field-simulation tables are labelled synthetic and were generated from simulated confidence timelines for three events. They are not pooled with the test results and are not used to estimate alarm rate, lead time, or fusion benefit.

3.2 Dataset composition and labels

D-Fire contains 21,527 images in the supplied copy. The documented partition has 15,063 training images, 3,200 validation images, and 3,264 test images. Across the complete dataset, the audit lists 14,692 fire box instances and 11,865 smoke box instances. The test set contains 3,990 annotated

object instances. For image-level evaluation, 1,504 test images are designated as background/negative scenes. These counts refer to different units: an image can contain zero, one, or multiple object boxes, and the number of images is not interchangeable with the number of instances.

The classes reported by the detector are fire and smoke. An image-level binary assessment is also calculated from whether the relevant target is detected in an image. This image-level assessment is useful for distinguishing positive from negative frames, but it does not assess the quality of each predicted bounding box. For example, a frame with multiple annotated instances can be counted as image-level positive if at least one instance is recovered even when other instances are missed. This distinction is central to interpreting the high image-level recall alongside the lower fire box recall.

FIgLib and a subset of FLAME are mentioned in the experimental inventory, but they do not contribute to the headline test results. The supplied FIgLib set has 732 images across ten sequences, while the FLAME 3 computer-vision subset has 2,214 RGB and thermal images. Neither is presented as an external test set for the reported YOLOv8n model. The available materials do not establish that the subset represents an independent geographic or temporal domain, nor do they provide a trained checkpoint evaluated consistently across all datasets. For that reason, these datasets are not used to support cross-dataset generalization claims.

3.3 Partition construction and leakage control

The dataset materials did not provide complete source-video identifiers. To reduce the risk that nearby frames would be randomly divided across the three partitions, the split-generation procedure assigned frames to numeric-stem buckets of 32 frames each. The audit reports 673 groups in total: 471 training groups, 100 validation groups, and 102 test groups. The separate verification file reports no group assigned to more than one split.

This design improves on a purely random frame split with respect to the constructed numeric-stem groups. It cannot establish that frames from the same source video or fire event are wholly confined to one split. The group definition is inferred from filenames rather than verified source metadata. If naming sequences restart, contain gaps, or combine different source videos, the buckets may not correspond exactly to independent events. The paper therefore uses the term *group-aware split* and explicitly identifies the group as a 32-frame numeric-stem bucket. It avoids describing the partition as a proven video-level split.

A stronger future partition should be created from immutable source-video or event identifiers, ideally accompanied by a manifest that records each image's original video, capture date, camera, location, and any near-duplicate relationship. The assignment should occur at the event level before image augmentation or frame extraction. A final external test set should remain untouched during model and threshold selection. Such a split would answer a more stringent generalization question than the current data permit.

3.4 Detector and training configuration

The evaluated model is a single-stream RGB YOLOv8n detector implemented with Ultralytics. It was trained for 150 epochs with an image size of 640 pixels and batch size 16. The optimizer was AdamW with an initial learning rate of 0.001 and a cosine schedule. The best checkpoint was selected at epoch 71 using validation mAP@0.5. The archive reports Python 3.11.9, PyTorch 2.14.1+cu130, torchvision 0.29.1+cu130, Ultralytics 8.4.170, and an NVIDIA GeForce RTX 5070 Laptop GPU with CUDA 13.0. The split-generation seed was 42. Fixed seeds support reproducibility, but they do not guarantee identical floating-point results across graphics drivers, hardware, or library implementations.

The model assessed here uses RGB imagery alone. The architecture diagram in the supplied archive depicts additional sensing and decision layers, including LWIR/thermal imagery, IIoT environmental measurements, a temporal contextual encoder, and alert fusion. Those depicted elements are not part of the measured detector checkpoint used for the principal results. The D-Fire development pathway included a grayscale proxy in a thermal stream, and the full dual-stream model was not trained to completion. These implementation details mean that the current results cannot quantify the incremental value of real thermal data, sensor fusion, temporal smoothing, or communication protocols.

3.5 Detection metrics

For a set of predicted detections and annotated objects, precision and recall are defined as

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \tag{1}$$

where TP, FP, and FN denote true-positive, false-positive, and false-negative object detections under the specified matching rule. The harmonic mean is

$$F_1 = 2\frac{PR}{P + R}. \tag{2}$$

These measures depend on the confidence threshold and the intersection-over-union (IoU) criterion used to match a predicted box to an annotation. Average precision (AP) summarizes the precision–recall curve for one class; mean AP (mAP) averages AP over classes. The supplied benchmark reports mAP at IoU 0.5 and averaged over IoU thresholds from 0.5 to 0.95. A stricter range of IoU thresholds penalizes imprecise localization more heavily than a single IoU threshold.

The supplied aggregate box-level metrics are reported at the benchmark's operating settings. Per-class precision, recall, F1, and AP are reported separately because an aggregate can conceal different error patterns between fire and smoke. No uncertainty interval is available for box-level metrics in the supplied summary, so point estimates are not accompanied by a newly estimated confidence interval. No paired significance test against published studies is performed because the same images, annotations, prediction outputs, and evaluation code are not available for those studies.

3.6 Image-level metrics

At the image level, each test image is assigned a binary status according to whether a target is present and whether the detector produces the corresponding detection. The supplied confusion matrix has 1,664 true-positive images, 96 false-negative images, 50 false-positive negative images, and 1,454 true-negative images. Thus, the image-level precision, recall, and accuracy are computed as

$$P_{img} = \frac{1664}{1664+50}, \qquad R_{img} = \frac{1664}{1664+96}, \\ A_{img} = \frac{1664+1454}{3264}. \tag{3}$$

The resulting values are approximately 0.971, 0.945, and 0.955. The project provides a 95% bootstrap interval of 0.948–0.962 for image-level accuracy. The archive does not provide sufficient detail in the summary to reconstruct every bootstrap implementation choice, such as the resampling code and random seed. The interval is therefore reported as supplied rather than described as an independently reproduced estimate.

Image-level outcomes should be read as frame-level presence detection. They do not measure how many objects were localized, how closely the predicted boxes align with ground truth, or how long an alert persists in consecutive frames. For an operational system, a second layer of event-level metrics would be required. Examples include event recall, time-to-detection, false alerts per camera-hour, alert duration, and the number of repeated alerts per incident. Such outcomes require timestamped video and an explicit event-matching protocol.

3.7 Confidence-threshold analysis

The confidence analysis includes 18 operating points in the supplied test-set sweep. The present paper highlights selected thresholds to illustrate the relationship between fire box recall and the proportion of negative images containing at least one fire detection. Let N_- be the number of negative test images and $n_{FP,img}(\tau)$ be the number of those images with at least one fire detection at confidence threshold τ. The negative-image false-positive rate is

$$FPR_{img}(\tau) = \frac{n_{FP,img}(\tau)}{N_-}, \quad N_- = 1504. \tag{4}$$

This quantity has image as its denominator. It does not account for repeated appearances of a scene over time and cannot be converted to false alarms per hour without additional temporal data.

The selected threshold points are descriptive operating choices. Threshold 0.15 favors recall more strongly; threshold 0.40 reduces the fraction of negative images with fire detections, while also lowering fire-instance recall. The sweep is performed on the held-out test split in the supplied artifacts. Consequently, the points should not be used to tune and then claim unbiased final-test performance. For a prospective deployment, a threshold should be selected on a validation set using a pre-specified cost or alert budget, then evaluated once on a separate, locked test set.

3.8 Latency measurement and reporting boundaries

The supplied operational metrics report 11.7 ms mean and 17.1 ms 95th-percentile latency per image. The timing was measured at FP32 with 640-pixel input on an RTX 5070 Laptop GPU and includes preprocessing and postprocessing. The raw latency file contains timing traces for 50 calls. These results characterize that software and hardware configuration only. They do not demonstrate end-to-end latency from camera exposure to operator notification, sustained throughput over continuous video, multi-camera scheduling, network transport, or performance under thermal and power constraints.

No Jetson Orin Nano INT8 inference result is available. No field false-alarm-per-hour metric is measured, since the archive lacks timestamped full-length negative footage. Those quantities are marked as not measured rather than estimated. This distinction is necessary because a projected inference speed, a synthetic replay, or a per-image throughput measurement is not equivalent to an observed field metric.

3.9 Evidence audit and reproducibility

Each numerical result in this paper is traceable to an experimental artifact in the supplied archive: the metrics JSON, benchmark CSV, per-class metrics, threshold-analysis CSV, operational metrics, data audit, leakage verification, training configuration, and latency traces. Synthetic field-simulation tables are excluded from empirical conclusions. The contextual transformer has a reported validation window accuracy of 0.936, but this is not a held-out end-to-end alert metric and is therefore not used as a principal outcome. Likewise, qualitative false-positive images are used to illustrate visible failure types, not to estimate their population frequency beyond the catalogued test subset.

4 Results

4.1 Test-set composition and aggregate results

The held-out D-Fire partition contains 3,264 images and 3,990 annotated instances. Table 1 presents the principal aggregate results. At the box level, the detector's mAP@0.5 is 0.719 and mAP@0.5:0.95 is 0.385. Aggregate precision is 0.739, recall is 0.664, and F1 is 0.700. At the image level, precision is 0.971, recall is 0.945, and F1 is 0.958. The image-level accuracy is 0.955, with the supplied interval of 0.948–0.962.

The difference between the box and image metrics is substantial. Image-level recall is 0.945, but aggregate box-level recall is 0.664 and fire-specific box recall is 0.572. This does not indicate a contradiction. The image-level measure asks whether a positive image was identified at least once, while box-level recall counts missed instances. A frame with several instances may therefore be a correct positive image even if one or more objects are missed. The difference shows why image-level accuracy alone is insufficient for evaluating localization performance.

TABLE 1
Measured performance on the D-Fire test split.

Metric	Box level	Image level
Precision	0.739	0.971
Recall	0.664	0.945
F1	0.700	0.958
mAP@0.5	0.719	–
mAP@0.5:0.95	0.385	–
Accuracy (95% CI)	–	0.955 [0.948, 0.962]

Test: 3,264 images, 3,990 object instances. The accuracy interval is the supplied 95% bootstrap interval.
Latency: 11.7 ms mean and 17.1 ms p95 per image (RTX 5070 Laptop GPU, FP32, 640 px).

TABLE 2
Class-specific box-level results on the test set.

Class	P	R	F1	AP_{50}	$AP_{50:95}$
Smoke	0.786	0.720	0.752	0.777	0.453
Fire	0.699	0.572	0.629	0.650	0.326

4.2 Class-specific performance

Table 2 reports the available class-level values. Smoke is detected more effectively than fire at the reported operating point. Smoke AP@0.5 is 0.777, compared with 0.650 for fire. At the more stringent AP@0.5:0.95 measure, the corresponding values are 0.453 and 0.326. Smoke recall is 0.720 and fire recall is 0.572. The lower fire result may be operationally important, but the supplied aggregate data do not identify its causes by object size, distance, illumination, or scene type. It would be inappropriate to attribute the gap to any one factor without stratified analysis.

The precision–recall curves and normalized confusion matrix provide additional views of test performance (Figs. 2 and 3). These graphics are descriptive summaries of the same test predictions; they are not independent replications. The training curves are included to show the reported learning trajectory and validation selection behavior (Fig. 1). The best checkpoint was selected on validation mAP@0.5 at epoch 71, while training continued to 150 epochs. This selection procedure avoids choosing the checkpoint directly from test performance, although the threshold sweep itself remains descriptive of the test partition.

4.3 Image-level confusion matrix

At the image level, 1,664 positive images were correctly identified, 96 positive images were missed, 50 negative images were classified as positive, and 1,454 negative images were correctly classified. The total is 3,264 test images. These counts yield image-level precision of 0.971 and recall of 0.945. The false-positive count of 50 refers to the binary image-level evaluation at the reported operating configuration; it should not be conflated with the number of fire-class bounding-box detections or with a false-alarm count over continuous video.

The normalized confusion matrix makes the imbalance between positive and negative classes more visible, but a confusion matrix remains threshold-specific. When the confidence threshold changes, the number of positive predictions and the associated precision/recall values change. The matrix therefore describes one operating point and should be interpreted together with the threshold sweep rather than as a complete summary of model behavior.

TABLE 3
Selected confidence-threshold results from the test-set sweep.

τ	Fire P	Fire R	Smoke R	FP images (%) [n]
0.15	0.511	0.698	0.781	1.8 [27]
0.30	0.674	0.592	0.733	0.8 [12]
0.40	0.756	0.528	0.696	0.47 [7]

4.4 Threshold-dependent fire and smoke outcomes

The selected test-set operating points are shown in Table 3. At a threshold of 0.15, fire recall is 0.698 and 27 of 1,504 negative images contain a fire detection, corresponding to 1.8%. At threshold 0.30, fire recall is 0.592 and 12 negative images contain fire detections, corresponding to 0.8%. At threshold 0.40, fire recall is 0.528 and 7 negative images contain fire detections, or 0.47%. Smoke recall at these points is 0.781, 0.733, and 0.696, respectively. Fire precision at thresholds 0.15, 0.30, and 0.40 is 0.511, 0.674, and 0.756.

The selected operating points show the expected direction of the trade-off in the supplied evaluation: a higher confidence threshold reduces the number of negative images with fire detections, but it also reduces fire recall. From threshold 0.15 to 0.40, negative-image fire detections decrease from 27 to 7, while fire recall decreases from 0.698 to 0.528. The 0.30 point lies between these extremes. These results do not establish which threshold is preferable in practice. The cost of missing an early fire, the cost of dispatching an unnecessary response, and the capacity to review alerts differ among organizations and must be specified before selecting an operating point.

The threshold-response curve is plotted in Fig. 4. Since the threshold sweep was computed on the test set, it should be regarded as an analysis of the observed test behavior. For future deployment evaluation, threshold selection should be performed on a separate validation partition, documented before testing, and held fixed on an external test set. This would avoid using the same test set both to choose an operating threshold and to report the resulting performance.

4.5 Qualitative false-positive observations

The project archive contains 50 negative scenes with one or more detections at a confidence threshold of 0.25. Its index and annotated examples indicate that the higher-confidence false positives include golden-hour clouds behind treelines classified as fire, overcast or storm skies classified as smoke, and vertical man-made structures such as a construction crane classified as fire. These examples illustrate perceptual ambiguity between target classes and background patterns. They do not establish the prevalence of these error types in all wildfire monitoring conditions.

A useful next step would be to annotate each false positive using a pre-defined taxonomy, such as cloud, fog, dust, steam, lighting/reflection, industrial plume, structure, or unknown. The taxonomy should be applied by more

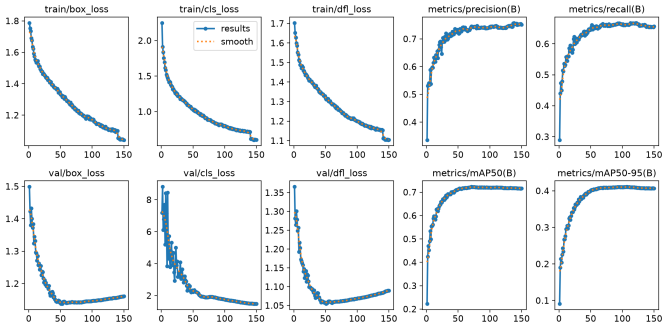


Fig. 1. Training and validation curves supplied with the experimental artifacts. The validation mAP@0.5 criterion selected the best checkpoint at epoch 71.

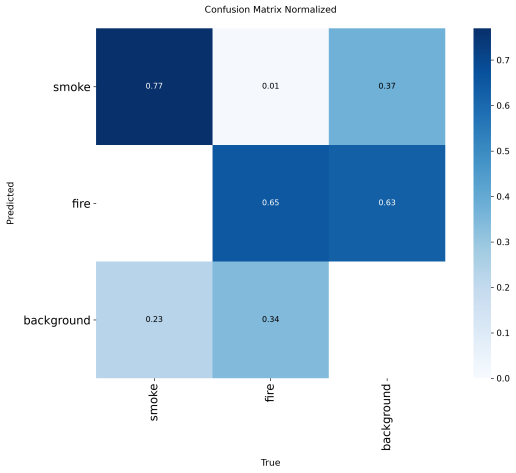


Fig. 2. Normalized confusion matrix on the held-out image test split.

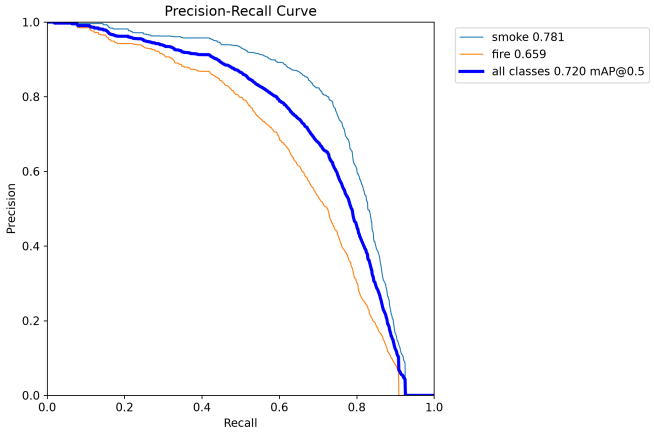


Fig. 3. Per-class precision–recall curves on the test split.

than one reviewer or supported by adjudication rules, and its agreement should be reported. False negatives should receive a parallel analysis based on object scale, contrast, occlusion, class, and illumination. Such analysis could explain which failure modes contribute to the observed fire–smoke difference. Those annotations were not part of the supplied experiment and are not presented as measured findings here.

4.6 Inference latency

The measured mean per-image latency is 11.7 ms, with a 95th percentile of 17.1 ms, for the specified RTX 5070 Laptop GPU, FP32 precision, and 640-pixel input. The timing includes preprocessing and postprocessing. This indicates that the evaluated software configuration processes individual test images within the reported latency on that device. It does not establish sustained throughput over long streams, queueing delay when several cameras are active, or the latency of a complete warning chain that includes camera acquisition, data transfer, temporal aggregation, message delivery, and human response.

No measurements are available for Jetson Orin Nano inference, INT8 quantization, power consumption, device temperature, or performance under constrained memory. The latency result is therefore hardware-specific and should not be generalized to edge devices. Similarly, the materials do not contain timestamped continuous negative video. False alarms per camera-hour cannot be estimated from the observed negative-image false-positive rate because the relationship depends on frame rate, temporal persistence, overlapping detections, and the algorithm used to consolidate repeated frame detections into one alert.

5 Discussion

5.1 Interpretation of image-level and object-level performance

The most important result is that image-level performance and object-level localization performance differ materially. Image-level recall of 0.945 indicates that most positive test images were recognized as containing a target according to the binary image-level rule. It does not imply that the detector found every annotated fire or smoke object. Fire box recall of 0.572 means that, under the box-matching criterion and operating configuration represented in the supplied metrics, a substantial proportion of annotated fire instances were not recovered. This distinction matters in a monitoring setting where missing a small or secondary flame may be important even if another target in the same image was detected.

For this reason, future evaluations should report at least two complementary levels of performance. The first level should describe localization and class recognition, with AP, precision, recall, and F1 by class and preferably by object size. The second should describe operational image or event behavior, including false-positive images, event-level detection, persistence, and time-to-detection. Neither layer replaces the other. Image-level scores are easier to interpret for alarm presence, while object-level scores reveal localization behavior and missed instances. Event-level metrics connect individual frames to how an operator experiences the alert stream.

The result also cautions against interpreting the overall box-level recall of 0.664 as the recall for each class. The class-specific values show that smoke recall is higher than fire recall. A single macro-average can be useful, but it may hide a class that is more consequential for the intended

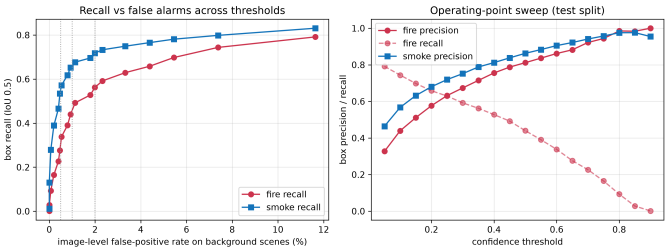


Fig. 4. Confidence-threshold analysis. Raising the threshold reduces the proportion of negative test images producing fire detections while lowering fire-instance recall.



Fig. 5. Qualitative prediction example supplied with the project. The image is illustrative and should not be interpreted as an independent performance measurement.

application. Reporting the separate fire and smoke values makes this difference visible. The available data do not show whether the difference arises from annotation characteristics, object scale, visual appearance, domain composition, or model capacity. A targeted error analysis is required before selecting an architectural remedy.

5.2 Threshold selection as an application decision

The threshold analysis provides a direct illustration of how confidence settings change a detector's operating profile. On this test partition, the 0.40 threshold gives the smallest selected negative-image false-positive proportion, 0.47%, but also the lowest selected fire recall, 0.528. The 0.15 threshold yields fire recall of 0.698, but 1.8% of negative images contain a fire detection. Neither point can be described as optimal without an explicit objective function or operational budget.

A fire-monitoring operator may prefer a lower threshold if missing an event has a much higher cost than reviewing a false alert. Another operator may face limited review staff or response resources and need a stricter alert budget. These preferences should be converted into a pre-specified threshold-selection rule rather than chosen after inspecting test results. Calibration should also be considered: a confidence score is not automatically a calibrated probability that the detection is correct. The archive does not provide a calibration study on a separate validation set. The reported sweep therefore describes the empirical threshold trade-off but does not establish probability calibration or expected utility under any particular cost structure.

The negative-image rate in this study is not a substitute for false alarms per hour. In continuous video, adjacent frames are correlated and the same object may produce detections over many consecutive frames. A temporal filter may merge these detections into one alert, or may suppress a genuine short-duration signal. The mapping from frame-level false detections to operator-facing alerts depends on the detection cadence, temporal smoothing, cooldown interval, camera uptime, and alert aggregation logic. A future study should define these choices explicitly and report both raw frame-level results and consolidated event-level results.

5.3 Dataset grouping and validity of the test result

Grouping every 32 numeric stems into one unit is a practical response to the absence of source-video identifiers in the supplied copy. It is preferable to ignoring the possibility of adjacent frames, and the verification file confirms that none of the constructed groups crosses the three partitions. However, the method cannot guarantee that one source event is not represented in multiple groups or that the 32-frame buckets align with actual video boundaries. The strength of the evidence is therefore intermediate: the partition is more conservative than a random image-level split with respect to the constructed groups, but less definitive than a source-video split.

This limitation affects how the result should be generalized. The reported metrics describe the held-out subset created by the documented procedure. They do not prove performance on a different region, camera network, season, or fire event. Even a perfectly event-separated test from the same benchmark would not necessarily establish geographic external validity. A rigorous follow-up should preserve an event-level test split and add an external set from a distinct source or deployment area. It should also report the rules for removing near-duplicates and the distribution of camera views and scene conditions across splits.

The threshold sweep also uses the held-out test data. The threshold values are presented here as an analysis of the dataset, but they should not be adopted as a final operational setting without an independent test after threshold selection. If a threshold is selected from a test curve and its corresponding recall is then presented as an unbiased estimate, the performance estimate may be optimistic. The validation partition is the appropriate place to choose the threshold, provided that the final test partition remains locked until the evaluation plan is fixed.

5.4 What the evidence supports about the proposed multimodal system

The supplementary architecture diagram describes RGB/PTZ cameras, thermal cameras, environmental sensors, a dual-stream detector, temporal encoding, evidence fusion, and message publication. Such a system may be a reasonable research direction, but a diagram is not an experimental result. The trained checkpoint measured in this paper is single-stream RGB. The grayscale proxy used in development is not equivalent to a registered LWIR measurement. The full dual-stream model was not trained, and the synthetic field-simulation tables contain only three simulated events. These facts preclude claims that thermal input improves detection, IIoT data improves precision,

temporal fusion reduces false alarms, or the full system detects events earlier than a camera-only baseline.

A valid multimodal comparison requires paired and temporally synchronized RGB and thermal observations from the same events, modality-specific quality controls, and a split that prevents the paired views of one event from appearing in different partitions. The baselines should include RGB-only, thermal-only, and fused models, with identical train/validation/test event assignments. The fusion mechanism and missing-modality behavior should be specified in advance. Results should include class metrics, calibration, event-level recall, alarm burden, latency, and robustness when one sensor is unavailable. Without these elements, the relative value of each additional sensor cannot be isolated.

A similar standard applies to the contextual tier. The archive reports a validation accuracy of 0.936 for window classification on embeddings derived from training frames. This is an intermediate model result. It does not establish end-to-end performance because the supplied materials do not report a held-out, timestamped sequence evaluation with event matching and alert aggregation. To evaluate the contextual layer, the detector and temporal model should be run as a complete pipeline on videos unseen during training. The evaluation should report whether temporal aggregation suppresses false detections without delaying or missing genuine events.

5.5 Research gaps and recommended experimental sequence

The findings identify several concrete gaps between the current benchmark and an operational wildfire-warning claim. First, the split lacks confirmed video and event identifiers. Second, the evaluation is limited to one benchmark distribution and one trained RGB checkpoint. Third, the test set contains images rather than continuous timestamped video for an alarm-rate calculation. Fourth, real paired thermal data and completed multimodal training are absent. Fifth, edge hardware and power characteristics were not measured. These gaps are independent: resolving one does not automatically resolve the others.

A practical follow-up can be organized into sequential stages. The first stage is data provenance. Each image should be linked to its source video, event, camera, timestamp, and geographic context where permitted. Near-duplicate detection can supplement provenance checks but should not replace them. The split should then be made at the event level, before augmentations, with an external test source reserved if available.

The second stage is protocol locking. The research team should define class and object-matching rules, confidence-threshold selection, image-to-event conversion, temporal persistence, and the treatment of ambiguous scenes. The threshold should be selected on a validation set according to a documented false-alert budget or cost ratio. The final test should remain untouched until model selection and threshold calibration are complete. Any change to the model or threshold after testing should be evaluated on a new test partition.

The third stage is model comparison. At minimum, an RGB-only baseline should be compared with the proposed enhanced model using identical data and training budgets. If thermal and environmental inputs are proposed, RGB-only, thermal-only, environmental-only, and fused variants should be evaluated. Ablations should remove one branch at a time, and the analysis should test both full-modality operation and realistic missing-sensor conditions. Reported gains should be accompanied by uncertainty estimates based on independent events rather than treating every correlated frame as an independent observation.

The fourth stage is operational evaluation. Continuous footage should include both fire-positive events and sufficiently long negative periods from relevant conditions. Outcomes should include event recall, time-to-detection measured from a pre-defined event onset, false alerts per camera-hour, alert duration, and missed-event review. Where multiple cameras are used, evaluation should include concurrency and network conditions. Performance should also be stratified by distance, object size, illumination, weather, camera movement, and confounding source when those labels are available.

The fifth stage is deployment testing. The model should be exported and benchmarked on the actual edge device. End-to-end latency, sustained throughput, peak memory, energy use, temperature, dropped frames, and recovery after network interruption should be recorded. Quantization should be treated as a new model configuration and evaluated for both speed and accuracy. These experiments would support a more complete deployment claim than single-image GPU inference latency.

5.6 Limitations

Several limitations constrain the interpretation of this paper. First, the 32-frame numeric-stem buckets are only a proxy for source-video or event separation. Second, all principal test results come from the D-Fire distribution; no external dataset test is reported. Third, the confidence-threshold sweep was performed on the test split in the supplied artifacts and must not be treated as an independently selected operating point. Fourth, the bootstrap interval is reported from the archive, but details needed to reproduce its resampling procedure are not fully specified in the summary materials. Fifth, the false-positive examples are a curated set of 50 scenes and do not constitute a complete error taxonomy.

Sixth, the measured latency is specific to an RTX 5070 Laptop GPU using FP32 and 640-pixel images. No embedded-device inference or power measurement is available. Seventh, alarm frequency per hour and event-level lead time are not measured because continuous timestamped video is unavailable. Eighth, the principal detector is RGB-only; thermal and IIoT branches in the system diagram do not have completed empirical results. Ninth, the synthetic replay results cannot be interpreted as field evidence. Finally, the absence of a matched baseline prevents a claim of comparative improvement. These limitations are not minor qualifications: they define the boundary of the claims that can be made from this dataset and experiment.

5.7 A prospective validation protocol

The evidence gaps identified above can be addressed through a staged protocol that separates data construction,

model development, operating-point selection, and final evaluation. This separation is important because each stage can otherwise influence the estimate produced by the next. For example, if the test partition is used to select a confidence threshold, then the final reported point is partly adapted to that test set. Similarly, if paired images from one event appear across training and test partitions, an external event may be less novel to the model than the partition names imply.

The first protocol stage should define the unit of independence. The recommended unit is a source event, or a source video when event linkage is unavailable. Each frame should retain a stable identifier connecting it to its original recording. A manifest should include the recording identifier, frame timestamp, camera identifier, and dataset partition. Any event-level grouping should be checked by a script that fails if one group appears in more than one split. Where a dataset contains several camera angles of the same burn, these views should be grouped together if the intended claim concerns generalization to unseen events. A separate camera-held-out test can answer the different question of generalization to new hardware or vantage points.

The second stage should define annotation and matching conventions. The project should state how partially visible flames and diffuse smoke are labelled, whether overlapping regions receive separate boxes, and how ambiguous background scenes are adjudicated. The IoU rule for a correct box match should be fixed before final evaluation. Reviewers should inspect a representative sample of both positive and negative labels, and disagreements should be documented. If the scientific question concerns early detection of a fire, the annotation protocol should distinguish visible smoke onset, visible flame onset, and the first frame at which an independent observer can identify an event. Without a predefined onset, a measured detection delay would depend on retrospective interpretation.

The third stage should reserve three roles for the data partitions. Training data should support parameter estimation and augmentation. Validation data should support model selection, confidence calibration, and temporal-policy design. The test data should be selected by event and should not be used to revise the model, threshold, persistence interval, or error taxonomy. An external test from another geography or capture campaign should be included where possible. If several candidate models are considered, the number of comparisons and the selection rule should be reported so that performance is not presented as though a single model had been chosen in advance.

The fourth stage should specify the operational objective. A decision threshold can be selected to maximize a metric such as F1, to satisfy a maximum false-alert budget, or to minimize an application-specific cost. These approaches answer different questions. A maximum-F1 threshold treats precision and recall symmetrically in the metric, whereas operational costs may be asymmetric. A false-alert budget requires a denominator and observation period. For a continuous stream, the protocol should define a minimum quiet interval or event-merging rule so that repeated detections from one plume are not automatically counted as many independent alerts. The selected policy should be finalized on validation footage and then applied unchanged to the held-out test footage.

The fifth stage should report outcomes at multiple levels. Frame-level detection can be summarized with precision, recall, F1, and AP by class. Event-level evaluation can report the proportion of annotated events detected within a prespecified time window, the distribution of time-to-detection, false alerts per camera-hour, and missed events. Alert-level evaluation should report the number and duration of alerts after temporal consolidation. Each metric should include its denominator and the unit of observation. Confidence intervals should be computed by resampling independent events or videos rather than treating adjacent frames as independent observations. If the number of independent events is small, that limitation should be presented alongside the interval rather than hidden by a large frame count.

The sixth stage should examine robustness and practical system constraints. The test protocol can stratify results by day and night, target size, distance, weather, camera motion, visibility, and types of confusing non-fire sources. These strata should be defined before error analysis to reduce selective reporting. If sensor fusion is proposed, paired inputs should be synchronized and evaluated with RGB-only, thermal-only, and fused variants using the same event splits. A modality-ablation experiment should remove one input branch at a time. The study should additionally report behavior when a sensor is absent, blocked, delayed, or degraded, because an operational system cannot assume that every sensor remains available.

The final stage should measure the complete deployment path. The intended camera and edge processor should be specified, and the model should be run continuously for a duration representative of use. Measurements should include per-frame inference latency, end-to-end alert latency, sustained throughput, peak memory, power consumption, temperature, dropped frames, storage requirements, and recovery after communication loss. If a quantized or accelerated model is evaluated, its accuracy and operating threshold should be re-measured rather than inherited from the FP32 model. A report should distinguish latency under a single-image benchmark from latency under the full stream-processing pipeline.

This staged protocol is not an outcome of the current experiments. It is a proposed design for a follow-up study, based on the limitations of the supplied archive. Its practical value is that it makes the evidence required for each claim explicit. A detector claim requires held-out detection results; an event-warning claim requires timestamped events and event-level scoring; a fusion claim requires paired modality experiments; and an edge-readiness claim requires measurements on the target device. Keeping those claims separate should make future results easier to reproduce and assess.

5.8 Implications for reporting and reproducibility

The present case illustrates why an experimental paper should include more than a single metric table. A compact reporting package should identify the model checkpoint, the data version, the image list for each split, the label convention, the matching rule, the confidence threshold, and the software and hardware versions. The metric output

should be regenerable from raw predictions and annotations. Where full datasets cannot be redistributed, stable identifiers and instructions for obtaining access can still help other researchers verify the evaluation. The supplied archive contains raw test predictions and a leakage-verification file, which are useful for such an audit, but the source-video identity limitation should remain visible.

Reproducibility also requires distinguishing repeated measurement from independent evidence. Multiple thresholds computed from one test set are not multiple independent test experiments. Several plots computed from the same predictions are alternative summaries, not independent replications. A bootstrap over images may estimate uncertainty conditional on the sampled image set, but if images are correlated within source video, image-level resampling can understate uncertainty about performance on new events. A future analysis should therefore resample at the group or event level once reliable grouping metadata is available.

The same discipline applies to synthetic evidence. Simulated confidence traces may help test whether a software implementation follows its specified logic. They cannot establish that the trace resembles a real camera stream or that the resulting system performs well in the field. The simulation's input assumptions, number of synthetic events, and purpose should be stated. In the supplied materials, the replay tables are marked synthetic; the current paper preserves that distinction and does not combine their results with the held-out image metrics.

Finally, a reproducibility statement should be precise about what is available and what remains restricted. The source project identifies dataset provenance and license-sensitive resources. Before publication, the authors should publish or provide controlled access to the split manifest, evaluation scripts, environment specification, and raw predictions where legally permissible. The model weights should be linked if redistribution is permitted. These actions would allow readers to check reported values without implying that every underlying image is freely redistributable.

6 Conclusion

This paper provides a detailed, evidence-bounded evaluation of a single-stream RGB YOLOv8n detector on a group-aware D-Fire test partition. The model achieves mAP@0.5 of 0.719 and mAP@0.5:0.95 of 0.385. Image-level accuracy is 0.955, but fire box recall is 0.572 at the reported operating point. The difference illustrates why image-level detection and object localization should be reported separately. The threshold analysis further shows that reducing the share of negative images with fire detections is accompanied by lower fire-instance recall.

The results are useful as benchmark evidence for the tested configuration, but they do not demonstrate field-ready early warning. The source-video grouping is approximate, external-domain performance is unknown, false alarms per camera-hour were not measured, and thermal fusion, full dual-stream training, edge-device operation, and system-level time-to-detection remain unvalidated. Future work should first secure event-level data provenance, then lock the evaluation protocol, test on independent events, and measure continuous-video and target-hardware performance. A cautious description of these boundaries is necessary for the results to be interpretable and reproducible.

Data and Code Availability

The data and code used in this study are available from the corresponding author upon request.

Funding and Conflicts of Interest

The authors declare that they have no conflicts of interest that could have influenced the work reported in this paper.

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